

LECTURE 4

Types of drug-receptor interactions, Types of adverse drug reactions

LEARNING OUTCOMES

By the end of this lecture, students should be able to:

Recognize drug binding forces

Define agonist and antagonist

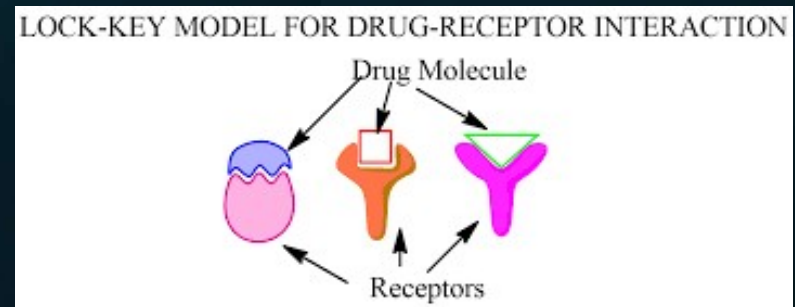
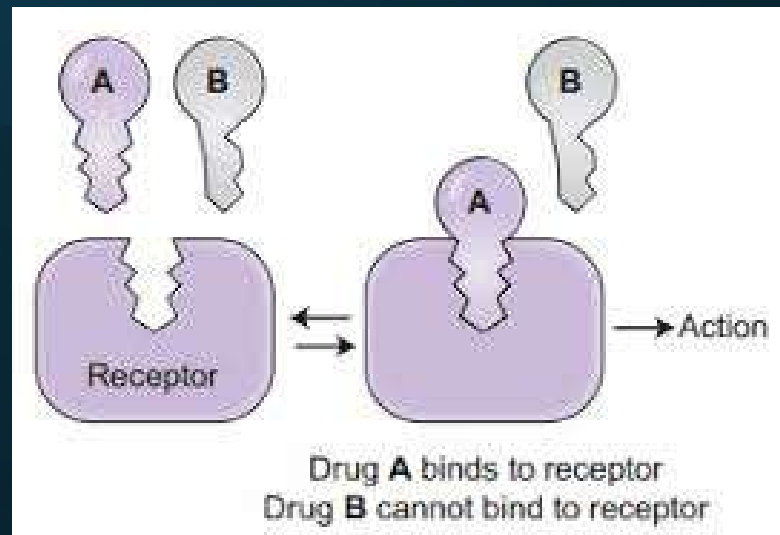
List types of adverse drug reactions

Explain precautions to minimize drug allergy

Drug-receptor Interactions

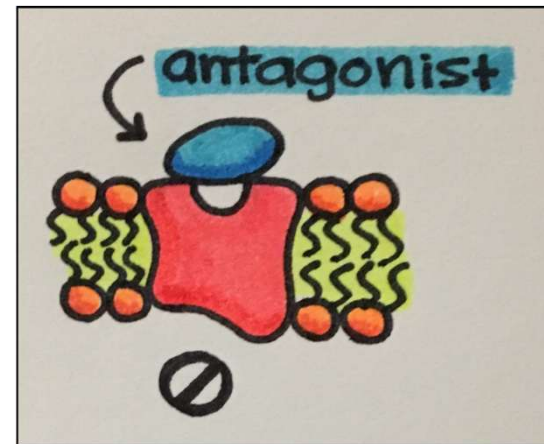
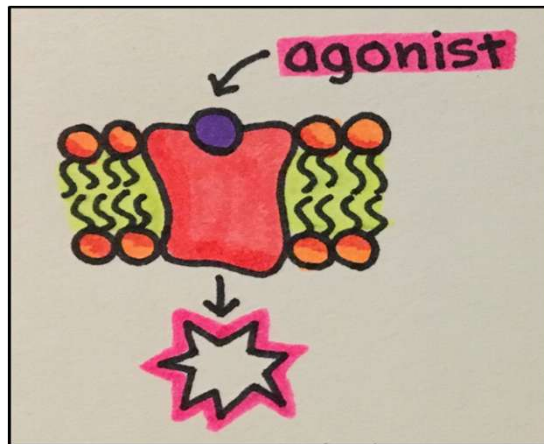
- **Receptors:** Protein molecules including enzymes, transporters and ion channels where a ligand (specific **endogenous** neurotransmitter/hormone or an external pharmacological agent (drug) binds to, resulting in a cellular response.
- **Unique Exception: Orphan Receptors** are receptors for which the ligand remains unknown.
- **Reminder: Ligand** is an ion or molecule that forms a complex with a biomolecule to serve a specific biological purpose
- Examples of **Endogenous Ligands:**
 - **Neurotransmitters[NT]:** chemical messengers signalling across a synaptic cleft
 - Acetylcholine [**Ach**]
 - Epinephrine [**EPI**]
 - Norepinephrine [**NE**]

DRUG-RECEPTOR INTERACTIONS



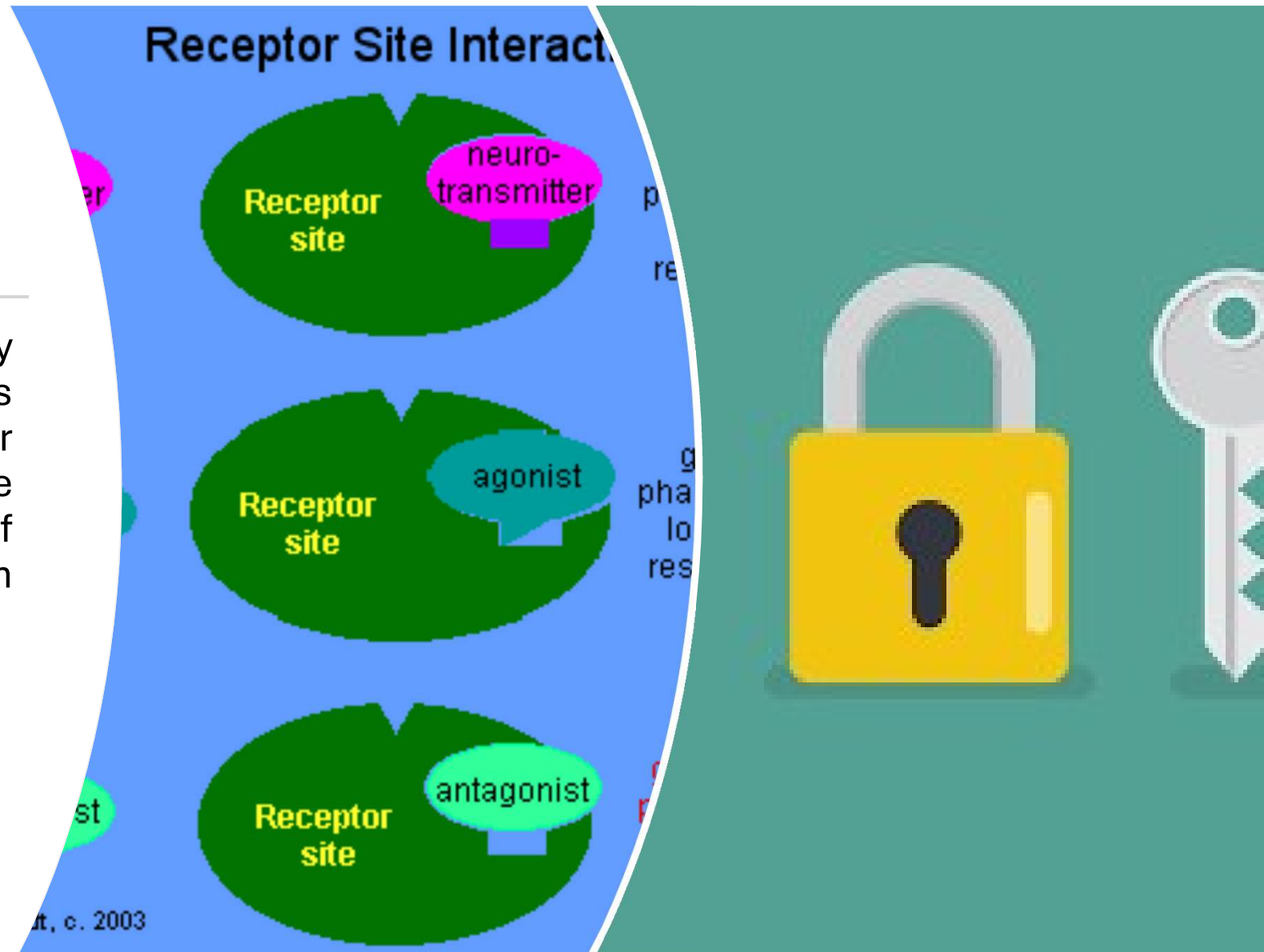
AGONISTS AND ANTAGONISTS

- **DRUG:** A chemical agent that selectively interacts with specific target molecules (*i.e.* receptors) to alter their specific physiological functions.
- **Agonist:** a drug that activates receptors to result in either stimulation or inhibition of the function of various types of cells and organs.
- **Antagonist:** a drug that prevents receptor activation by agonists.



Receptor Interactions

- The lock and key metaphor is useful for understanding the interaction of receptors with their ligands.



DRUG-RECEPTOR BINDING

- **Drugs** bind to their respective receptor in a variety of ways depending on their characteristics.
 - A. Ionic interaction:** cation & anion
 - B. Hydrogen bonding**
 - C. Lipophilic interaction**
 - D. Covalent bond:** irreversible



ADVERSE DRUG REACTIONS

Adverse drug reaction is an undesired or unintended effect of the drug, occurs at dose normally used by human being.

The adverse drug reaction requires treatment or decrease in dose if it is due to poisoning or overdose.

Occurs at doses of an appropriately given drug used in man for prophylaxis, diagnosis or therapy excluding therapeutic failures.

TYPES OF ADVERSE DRUG REACTIONS

TYPE	TERM	CHARACTERISTICS	EXAMPLES
A	Augmented	Dose-dependent, frequent, predictable, drug overdose, explained by pharmacologic drug effect	Bleeding after anticoagulants, hypoglycemia from insulin
B	Bizarre	Dose-independent, rare, unpredictable, not explained by pharmacologic drug effect	Urticaria from aspirin, exanthem from antibiotics
C	Chronic	Dose and time dependent, rare, long-term exposure	Cushing syndrome from cortisone
D	Delayed	Time dependent, very rare	Kidney disease from long-term analgesics or non-steroidal anti-inflammatory drugs (NSAIDs)
E	End of treatment	Time dependent, rare, relapse after stopping a therapy	Withdrawal effects of drugs (eg opioids or antiepileptics), rebound or relapse phenomena

DRUG ALLERGY

DRUG ALLERGY IS A CELL-MEDIATED IMMUNE REACTION PRODUCING SYMPTOMS UNRELATED TO THE DRUG'S PHARMACOLOGICAL EFFECTS.

DUE TO PREVIOUS SENSITIZATION TO THE DRUG ITSELF OR A PARTICULAR CHEMICAL WITH A SIMILAR STRUCTURE. PRIOR SENSITIZATION IS NEEDED, AND A LATENT PERIOD OF AT LEAST 1-2 WEEKS IS REQUIRED AFTER THE FIRST EXPOSURE.

ADMINISTER OXYGEN AT HIGH FLOW RATE AND ADRENALINE(0.5 MG IM) INJECTION.

ADMINISTER AN H1 ANTIHISTAMINIC (CHLORPHENIRAMINE 10-20 MG) I.M./SLOW I.V.

INTRAVENOUS GLUCOCORTICOID (HYDROCORTISONE SOD. SUCCINATE-200 MG)

References

- Bart –Jhonson, Frank J. Dowd. ***Pharmacology and Therapeutic for Dentistry***, 6th edition, 2011. Elsevier Publishers,USA
- Karen Whalen, Richard Finkel, Thomas A Panavelil. **Lippincott Illustrated Reviews Pharmacology**. 6th ed. 2015. Philadelphia Wolter Kluwer Puplisher.

Thank You